

BRAMBLE–PASCIAK-TYPE CONJUGATE GRADIENT METHOD FOR DARCY’S EQUATIONS*

FAISAL FAIRAG[†], MOHAMMED ALSHAHRANI[†], AND HATTAN TAWFIQ[†]

Abstract. We consider the solution of a system of linear algebraic equations which is obtained from a Raviart–Thomas mixed finite element formulation of Darcy’s equations. In [C. E. Powell and D. Silvester, *SIAM J. Matrix Anal. Appl.*, 25 (2003), pp. 718–738], Powell and Silvester developed a block-diagonal preconditioner for this system. In this research work, we extend their results by constructing a block-triangular preconditioner and establish an eigenvalue bound for the preconditioned matrix. The preconditioned matrix is nonsymmetric but it is self-adjoint in a nonstandard inner product. The Bramble–Pasciak-type conjugate gradient method is used to solve the linear system which ensures that the norm of the error is minimized. Numerical experiments confirm the theoretical results and illustrate good convergence properties.

Key words. preconditioning technique, saddle point problems, Bramble–Pasciak CG method, $H(\text{div})$ preconditioner, Krylov subspace method, Darcy’s law

AMS subject classifications. 65F10, 65N22, 65M60, 65N12

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1. Introduction. We consider the problem of Darcy flow which describes the fluid flow in porous media. This problem is of considerable importance in many fields, including oil recovery and groundwater pollution modeling. We begin with the mixed formulation of Darcy flow

$$(1) \quad \begin{aligned} K^{-1}u - \nabla p &= 0, \\ \nabla \cdot u &= -f \quad \text{in } \Omega \end{aligned}$$

with some suitable boundary conditions, where Ω is a bounded domain in $\mathbb{R}^2$. The unknowns are the velocity u and the pressure p , while K represents the permeability. When Raviart–Thomas finite elements [23] are used, a sparse, indefinite, and ill-conditioned linear system is generated for (1). This linear system is known [1, 5, 6, 23, 24, 25] to be a saddle point problem of the form

$$(2) \quad \mathcal{A}x \equiv \begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} g \\ h \end{bmatrix} \equiv \mathbf{b}$$

with $x = [u \ p]^T$ and where $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite and $B \in \mathbb{R}^{m \times n}$ ($m \leq n$) is of full rank, g and h are vectors of sizes n and m , respectively. The saddle point matrix $\mathcal{A} \in \mathbb{R}^{(n+m) \times (n+m)}$ is symmetric, indefinite nonsingular and therefore the linear system (2) has a unique solution. However it is ill-conditioned (see for example [24]), and hence Krylov subspace methods are very slow and preconditioners are needed.

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[†]Mathematics and Statistics Department, King Fahd University of Petroleum and Minerals, Dhahran, 31261 Saudi Arabia (ffairag@kfupm.edu.sa, mshahrani@kfupm.edu.sa, hattan@kfupm.edu.sa).

Powell and Silvester [22] developed a block-diagonal preconditioner (P_{div}),

$$(3) \quad \begin{bmatrix} A+D & 0 \\ 0 & N \end{bmatrix}, \quad D \in \mathbb{R}^{n \times n}, N \in \mathbb{R}^{m \times m},$$

for this linear system and showed that the eigenvalues of the preconditioned matrix are contained in a union of two intervals

$$(4) \quad [-b, -a] \cup [1],$$

where the first interval is of small size and therefore preconditioned MINRES (PMINRES) is applicable and fast.

In this research work, we extend their results by constructing a block-triangular preconditioner,

$$(5) \quad \begin{bmatrix} \gamma_1 A + \gamma_2 D & 0 \\ B & -\alpha N \end{bmatrix}, \quad \gamma_1, \gamma_2, \alpha > 0,$$

and show that the eigenvalues of the preconditioned matrix are contained in a union of two intervals

$$(6) \quad [a_1, b_1] \cup [c] \quad a_1, b_1, c > 0,$$

where the first interval is of small size. Furthermore, the preconditioned matrix using the proposed preconditioner is both self-adjoint and positive definite in a nonstandard inner product. This enables us to apply the Bramble–Pasciak-type preconditioned conjugate gradient (CG) method where the norm of the error is minimized. This norm is equivalent to the Euclidean norm.

In 1988, Bramble and Pasciak [4] developed a block triangular preconditioner for the discrete Stokes problem. The beauty of their work lies in the fact that the original indefinite symmetric problem is turned into a self-adjoint and positive definite problem in a nonstandard inner product. Then, the well-known CG method [10], which is widely used for the solution of symmetric definite matrix systems, could be used in a nonstandard inner product.

In other words, the converted system is symmetric and positive definite in this inner product hence the CG method based on this inner product can be applied. There are many solvers [3, 7, 8, 12, 13, 14, 26, 27, 28, 31] of Bramble–Pasciak-type where a matrix is preconditioned and then a nonstandard inner product is used to solve the linear system by efficient Krylov subspace solvers like CG. For a more extensive survey of these solvers and other techniques, see [2, 20, 28]. For similar block-diagonal preconditioners for general saddle point problems one may consult [15, 25, 29].

The rest of the paper is organized as follows. The necessary preliminaries and background needed are introduced in section 2. In section 3, we discuss the properties of the preconditioned system using (5) showing that it is self-adjoint and positive definite in a nonstandard inner product. We also provide necessary conditions on the parameters of (5) under which the preconditioned matrix is positive definite. Section 4 discusses some spectral properties of the preconditioned matrix using (5). The modified CG method is presented in section 5. Numerical experiments demonstrating the spectral properties of the preconditioned system and illustrating the convergence of the preconditioned CG method in a nonstandard inner product with comparisons to the (P_{div}) preconditioner are given in section 6.

2. Preliminaries. In what follows, Ω is taken to be a convex polygon in $\mathbb{R}^2$ and $\partial\Omega = \partial\Omega_D \cup \partial\Omega_N$ is its boundary. Here $\partial\Omega_D$ and $\partial\Omega_N$ are the boundaries on which we have Dirichlet and Neumann boundary conditions, respectively. The space of square-integrable single-valued functions defined on Ω with the traditional inner product, $(\cdot, \cdot)$, is denoted by $L^2(\Omega)$. The space $L^2(\Omega)^2 = L^2(\Omega) \times L^2(\Omega)$ is the space of all vector functions whose components belong to $L^2(\Omega)$. For both spaces $\|\cdot\|_0$ denotes the norm. We define the two subspaces $H^{\text{div}}(\Omega)$ and $H^1(\Omega)$ as follows:

$$\begin{aligned} H^{\text{div}}(\Omega) &= \{v \in L^2(\Omega)^2 \mid \nabla \cdot v \in L^2(\Omega)\}, \\ H^1(\Omega) &= \{w \in L^2(\Omega) \mid \nabla w \in L^2(\Omega)^2\}. \end{aligned}$$

The inner products of $H^{\text{div}}(\Omega)$ and $H^1(\Omega)$ are, respectively,

$$\begin{aligned} (u, v)_{\text{div}} &= (u, v) + (\nabla \cdot u, \nabla \cdot v), \\ (u, w)_1 &= (u, w) + (\nabla u, \nabla w), \end{aligned}$$

and $\|\cdot\|_{\text{div}}$ and $\|\cdot\|_1$ are the induced norms. We also denote by $H^{\frac{1}{2}}(\partial\Omega_D)$ the set of traces of $H^1(\Omega)$ functions on $\partial\Omega_D$.

Let us now consider the first order system

$$\begin{aligned} (7) \quad & K^{-1}u - \nabla p = 0, \\ & \nabla \cdot u = -f \quad \text{in } \Omega, \\ & p = g \quad \text{on } \partial\Omega_D, \\ & u \cdot n = 0 \quad \text{on } \partial\Omega_N, \end{aligned}$$

where $f \in L^2(\Omega)$ and $g \in H^{\frac{1}{2}}(\partial\Omega_D)$ and n denotes the outward unit normal vector on $\partial\Omega_N$. In (7), $\partial\Omega_D \neq \emptyset$ and K is a 2×2 matrix-valued function. In this work, K is assumed to be bounded, symmetric, and uniformly positive definite. These properties ensure the existence of two positive constants $\hat{\gamma}$ and Γ with $0 < \hat{\gamma} < \Gamma$ such that

$$(8) \quad \hat{\gamma}(v, v) \leq (K^{-1}v, v) \leq \Gamma(v, v)$$

for every $v : \Omega \rightarrow \mathbb{R}^2$. First, we formulate (7) as a saddle point problem by choosing two test spaces,

$$(9) \quad V = H_{0,N}^{\text{div}}(\Omega) = \{v \in H^{\text{div}}(\Omega) \mid v \cdot n = 0 \text{ on } \partial\Omega_N\},$$

and $W = L^2(\Omega)$. Multiplying by arbitrary v and w in (7), applying Green's theorem in the first equation and imposing essential boundary conditions yield the variational problem, find $(u, p) \in V \times W$ satisfying

$$\begin{aligned} (10) \quad & (K^{-1}u, v) + (p, \nabla \cdot v) = \langle g, v \cdot n \rangle \quad \forall v \in V, \\ & (w, \nabla \cdot u) = -(f, w) \quad \forall w \in W, \end{aligned}$$

where $\langle g, v \cdot n \rangle = \int_{\partial\Omega_D} gv \cdot n ds$. Inequality (8) guarantees that the bilinear form $(K^{-1}u, v)$ is bounded and coercive. The stability theory of Brezzi and Babuska (see [6]) tells us that a unique solution (u, p) exists if and only if there are constants $\hat{\alpha}, \beta$, bounded away from zero, such that

$$(11) \quad (K^{-1}v, v) \geq \hat{\alpha} \|v\|_{\text{div}}^2 \quad \forall v \in V,$$

$$(12) \quad \sup_{0 \neq v \in V} \frac{(w, \nabla \cdot v)}{\|v\|_{\text{div}}} \geq \beta \|w\|_0 \quad \forall w \in W,$$

where $Z = \{v \in V \mid (w, \nabla \cdot v) = 0 \quad \forall w \in W\}$. We refer to the above conditions as coercivity and inf-sup stability, respectively. Since $\nabla \cdot V \subset W$, the subspace Z of V contains only vector functions with divergence-free velocities and the two norms $\|\cdot\|_{\text{div}}$ and $\|\cdot\|_0$ are equivalent on this subspace.

2.1. Discrete finite element spaces. Let T_h denote a triangulation of Ω into a mesh of triangles or rectangles $\{T_1, \dots, T_s\}$ with maximum edge size h . Let $\{T_{h1}, T_{h2}, \dots\}$ be a family of shape-regular, affine triangulations of Ω . All our numerical computations are performed on rectangular grids and so, for given h and T_h , the Raviart–Thomas spaces $V_h \subset H^{\text{div}}(\Omega)$ and $W_h \subset L^2(\omega)$ of integer index k (see [23, 22]) are,

$$(13) \quad V_h = \{v_h \in H^{\text{div}}(\Omega) \mid v_h|_T \in Q_{k+1,k}(T) \times Q_{k,k+1}(T) \quad \forall T \in T_h\},$$

and

$$(14) \quad W_h = \{w_h \in L^2(\Omega) \mid w_h|_T \in P_k(T) \quad \forall T \in T_h\}.$$

Here $P_k(T)$ is the space of polynomials of degree less than or equal to k on T and $Q_{r,s}(T)$ is the set of all polynomials of degree less than or equal to r in x and less than or equal to s in y . We restrict our discussion to the lowest order Raviart–Thomas space ($k = 0$). Velocities and pressures then have the special piecewise forms

$$(15) \quad v_h|_T = \begin{pmatrix} v_1 + v_2 x \\ v_3 + v_4 y \end{pmatrix}, \quad w_h|_T = w_1,$$

respectively, where v_1, v_2, v_3, v_4 , and w_1 are constants.

We start the Raviart–Thomas finite element discretization by choosing $V_h \subset V$ and $W_h \subset W$ defined above. Solving (10) over these spaces leads to the discrete problem, find $(u_h, p_h) \in V_h \times W_h$ satisfying

$$(16) \quad \begin{aligned} (K^{-1}u_h, v_h) + (p_h, \nabla \cdot v_h) &= \langle g, v_h \cdot n \rangle \quad \forall v_h \in V_h, \\ (w_h, \nabla \cdot u_h) &= -(f, w_h) \quad \forall w_h \in W_h. \end{aligned}$$

Stability of (16) has been given in [23]. The importance of selecting this combination of two discrete spaces is that the discrete inf-sup condition is satisfied.

To express (16) as a linear algebra problem, we choose bases such that $V_h = \text{span}\{\psi_i\}$ and $W_h = \text{span}\{\phi_i\}$. Defining $A \in R^{n \times n}$ by $A_{i,j} = (K^{-1}\psi_i, \psi_j)$ for $1 \leq i, j \leq n$, $B \in R^{m \times n}$ by $B_{k,j} = (\phi_k, \nabla \cdot \psi_j)$ for $1 \leq k \leq m$, and the vectors $f \in R^m, g \in R^n$ by $f_k = -(f, \phi_k)$ and $g_i = (g, \phi_i \cdot n)$ gives the problem

$$(17) \quad \begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} g \\ h \end{bmatrix}$$

with $u_h = \sum_{i=1}^n u_i \psi_i$, $p_h = \sum_{j=1}^m p_j \phi_j$, and $u = [u_1, u_2, \dots, u_n]^T$, $p = [p_1, p_2, \dots, p_m]^T$.

Next we define the velocity mass matrix $M \in R^{n \times n}$, the velocity divergence matrix $D \in R^{n \times n}$, and the pressure mass matrix $N \in R^{m \times m}$ by

$$(18) \quad M_{i,j} = (\psi_i, \psi_j), \quad 1 \leq i, j \leq n,$$

$$(19) \quad D_{i,j} = (\nabla \cdot \psi_i, \nabla \cdot \psi_j), \quad 1 \leq i, j \leq n,$$

$$(20) \quad N_{r,s} = (\phi_r, \phi_s), \quad 1 \leq r, s \leq m.$$

These matrices provide useful discrete representations of the norms $\|\cdot\|_0$ and $\|\cdot\|_{\text{div}}$ on V_h and the norm $\|\cdot\|_0$ on W_h since for any v_h in V_h and w_h in W_h we can write

$$(21) \quad \|v_h\|_0^2 = v^T M v,$$

$$(22) \quad \|v_h\|_{\text{div}}^2 = v^T (M + D) v,$$

$$(23) \quad \|w_h\|_0^2 = w^T N w,$$

where the vectors $v = [v_1, v_2, \dots, v_n]^T$ and $w = [w_1, w_2, \dots, w_m]^T$ are given as

$$v_h = \sum_{i=1}^n v_i \psi_i,$$

$$w_h = \sum_{j=1}^m w_j \phi_j.$$

It is clear that for the lowest-order Raviart–Thomas spaces, D and N are cheap to assemble. The entries are constants and no numerical integration is required. The matrix N is diagonal with entries $|T_1|, \dots, |T_m|$, where $|T_i|$ denotes the area of the i th rectangle. Notice also that for unit diffusion coefficients $K = I$ (the identity matrix), the mass matrix M is identical to A . The three matrices B, D , and N are related and the relationship between these matrices is given in the following lemma. This lemma and its proof are available in [22].

LEMMA 2.1. *If $\nabla \cdot V_h \subset W_h$, then,*

$$(24) \quad D = B^T N^{-1} B.$$

By writing the inf-sup condition (12) in discrete form we see that the inf-sup constant β satisfies,

$$(25) \quad \beta^2 \leq \frac{w^T B(M + D)^{-1} B^T w}{w^T N w}, \quad \forall w \in R^m,$$

and so β is $\sqrt{\lambda_{\min}}$, where $\lambda_{\min}$ is the minimum eigenvalue of $N^{-1} B(M + D)^{-1} B^T$. Note that β does not depend on K and so the method is inf-sup stable even when $\frac{\Gamma}{\gamma}$ is large. By Cauchy–Schwarz we also have

$$(26) \quad \langle w_h, \nabla \cdot v_h \rangle \leq \|w_h\|_0 \|\nabla \cdot v_h\|_0 \leq \|w_h\|_0 \|v_h\|_{\text{div}} \quad \forall v_h \in V_h, \quad w_h \in W_h.$$

and therefore

$$\beta \leq 1.$$

Powell and Silvester [22] used the block-diagonal matrix,

$$(27) \quad P_{\text{div}} = \begin{bmatrix} M + D & 0 \\ 0 & N \end{bmatrix},$$

to precondition the linear system (2). They call P_{div} an “ $H(\text{div})$ ” preconditioner since for any v_h in V_h ,

$$(28) \quad v^T \left(\frac{1}{\Gamma} M + D \right) v \leq \|v_h\|_{\text{div}}^2 \leq v^T \left(\frac{1}{\gamma} M + D \right) v,$$

where γ and Γ are the constants that describe the boundedness of the diffusion coefficients in (8). That is $M + D$ gives a discrete representation of a norm that is equivalent to $\|\cdot\|_{\text{div}}$ on V_h .

3. Proposed preconditioner. In this section, based on the preconditioner introduced by Powell and Silvester in [22], we propose a new block-triangular preconditioner defined as

$$(29) \quad P(\gamma_1, \gamma_2, \alpha) = \begin{bmatrix} \gamma_1 A + \gamma_2 D & 0 \\ B & -\alpha N \end{bmatrix},$$

where γ_1, γ_2 , and α are positive scalars. The preconditioned system, therefore, is

$$(30) \quad \tilde{\mathcal{A}}x \equiv P^{-1}(\gamma_1, \gamma_2, \alpha)\mathcal{A}x = P^{-1}(\gamma_1, \gamma_2, \alpha)\mathbf{b} \equiv \tilde{\mathbf{b}}.$$

It is easy to see that the preconditioned matrix $\tilde{\mathcal{A}}$ is self-adjoint with respect to the inner product defined as $\langle x, y \rangle_{\mathcal{H}} = y^T \mathcal{H}x$ for all $x, y \in \mathbb{R}^n$ with $\mathcal{H}$ given as

$$(31) \quad \mathcal{H} = \begin{bmatrix} (1 - \gamma_1)A - \gamma_2 D & 0 \\ 0 & \alpha N \end{bmatrix};$$

that is, $\tilde{\mathcal{A}}^T \mathcal{H} = \mathcal{H} \tilde{\mathcal{A}}$. It should be noted that $\mathcal{H}$ is symmetric and positive definite under the assumption that $(1 - \gamma_1)A - \gamma_2 D$ is positive definite for some positive scalars γ_1 and γ_2 . Also, it can be shown that $\tilde{\mathcal{A}}$ is positive definite with respect to the same inner product, in the sense that $\langle \tilde{\mathcal{A}}x, x \rangle_{\mathcal{H}} > 0$ for all $0 \neq x \in \mathbb{R}^n$; in fact, as in [28], we may factorize $\tilde{\mathcal{A}}\mathcal{H}$ as

$$(32) \quad \begin{aligned} \tilde{\mathcal{A}}^T \mathcal{H} &= \begin{bmatrix} A(\gamma_1 A + \gamma_2 D)^{-1}A - A & A(\gamma_1 A + \gamma_2 D)^{-1}B^T - B^T \\ B(\gamma_1 A + \gamma_2 D)^{-1}A - B & B(\gamma_1 A + \gamma_2 D)^{-1}B^T \end{bmatrix} \\ &= \begin{bmatrix} I & 0 \\ BA^{-1} & I \end{bmatrix} \begin{bmatrix} A(\gamma_1 A + \gamma_2 D)^{-1}A - A & 0 \\ 0 & BA^{-1}B^T \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix}, \end{aligned}$$

which is a congruence transformation. The number of positive eigenvalues and the number of negative eigenvalues of the matrix $\tilde{\mathcal{A}}^T \mathcal{H}$ are determined by the eigenvalues of the matrix

$$\begin{bmatrix} A(\gamma_1 A + \gamma_2 D)^{-1}A - A & 0 \\ 0 & BA^{-1}B^T \end{bmatrix}.$$

This is due to Sylvester's law of inertia. Since $BA^{-1}B^T$ is positive definite, the matrix $\tilde{\mathcal{A}}\mathcal{H}$ becomes positive definite if the matrix $A(\gamma_1 A + \gamma_2 D)^{-1}A - A$ is positive definite. We can rewrite this matrix as $A((\gamma_1 A + \gamma_2 D)^{-1} - A^{-1})A$, which will be positive definite if the matrix $(\gamma_1 A + \gamma_2 D)^{-1} - A^{-1}$ is positive definite. This is equivalent to

$$(33) \quad y^T(\gamma_1 A + \gamma_2 D)y < y^T A y.$$

Also, it is equivalent to

$$(34) \quad y^T((1 - \gamma_1)A - \gamma_2 D)y > 0.$$

The condition (34) is what is needed to ensure the positive definiteness of H .

LEMMA 3.1. *There exist two positive numbers β_1 and β_2 such that*

$$\beta_1(Dx, x) \leq \beta_2(Ax, x) \quad \text{for all } x \in \mathbb{R}^n.$$

Proof. Let $\lambda_{\max}(D)$ denote the largest eigenvalue of D and v its associated eigenvector and $\lambda_{\min}(A)$ the smallest eigenvalue of A . Since D is positive semidefinite and A is positive definite, let β_1 and β_2 be two positive numbers such that

$$\beta_1 \lambda_{\max}(D) \leq \beta_2 \lambda_{\min}(A).$$

Now let $0 \neq x \in \mathbb{R}^n$, then

$$\begin{aligned}\beta_1(Dx, x) &= x^T x \beta_1 \frac{x^T Dx}{x^T x} \leq x^T x \beta_1 \lambda_{\max}(D) \leq x^T x \beta_2 \lambda_{\min}(A) \\ &= x^T x \beta_2 \min_{0 \neq y \in \mathbb{R}^n} \frac{y^T Ay}{y^T y} \leq \beta_2 x^T Ax = \beta_2(Ax, x),\end{aligned}$$

as desired. $\square$

LEMMA 3.2. *There exist two positive numbers γ_1 and γ_2 such that*

$$(35) \quad \alpha_0(Ax, x) \leq ((\gamma_1 A + \gamma_2 D)x, x) \leq \alpha_1(Ax, x)$$

for some positive numbers $\alpha_0 < 1$ and $\alpha_1 < 1$ and for all $x \in \mathbb{R}^n$.

Proof. By Lemma 3.1, there exist two positive numbers β_1 and β_2 such that

$$\beta_1 x^T Dx \leq \beta_2 x^T Ax \quad \text{for all } x \in \mathbb{R}^n.$$

Then

$$\tau x^T Ax + \beta_1 x^T Dx \leq \tau x^T Ax + \beta_2 x^T Ax \quad \text{for all } x \in \mathbb{R}^n \quad \text{and all } \tau > 0.$$

From the positive semidefiniteness of the matrix D , we get

$$\tau x^T Ax \leq \tau x^T Ax + \beta_1 x^T Dx \leq (\tau + \beta_2) x^T Ax \quad \text{for all } x \in \mathbb{R}^n \quad \text{and all } \tau > 0.$$

Now let μ be any positive number and divide the last inequality by $\tau + \beta_2 + \mu$; that is,

$$\begin{aligned}\frac{\tau}{\tau + \beta_2 + \mu} x^T Ax &\leq \frac{\tau}{\tau + \beta_2 + \mu} x^T Ax + \frac{\beta_1}{\tau + \beta_2 + \mu} x^T Dx \\ &\leq \frac{\tau + \beta_2}{\tau + \beta_2 + \mu} x^T Ax \quad \text{for all } x \in \mathbb{R}^n.\end{aligned}$$

We conclude the proof by setting

$$\begin{aligned}\gamma_1 &= \frac{\tau}{\tau + \beta_2 + \mu}, \quad \gamma_2 = \frac{\beta_1}{\tau + \beta_2 + \mu}, \\ \alpha_0 &= \frac{\tau}{\tau + \beta_2 + \mu}, \quad \text{and } \alpha_1 = \frac{\tau + \beta_2}{\tau + \beta_2 + \mu}.\end{aligned} \quad \square$$

Remark 3.1. Using Lemma 3.2, we can see that

$$0 < (1 - \alpha_1)\langle Ax, x \rangle \leq \langle (1 - \gamma_1)A - \gamma_2 D, x \rangle \leq (1 - \alpha_0)\langle Ax, x \rangle \quad \text{for all } 0 \neq x \in \mathbb{R}^n.$$

The above inequality ensures that all eigenvalues of the matrix $(1 - \gamma_1)A - \gamma_2 D$ are positive. This ensures that $\mathcal{H}$ is positive definite and hence defines an inner product.

Remark 3.2. In order for matrix $(1 - \gamma_1)A - \gamma_2 D$ to be positive definite, we must have

$$\frac{1 - \gamma_1}{\gamma_2} > \frac{\langle Dx, x \rangle}{\langle Ax, x \rangle} \quad \text{for all } x \in \mathbb{R}^n.$$

Taking the maximum over all $0 \neq x \in \mathbb{R}^n$, we have the following lower bound of the ratio $\frac{1 - \gamma_1}{\gamma_2}$,

$$(36) \quad \frac{1 - \gamma_1}{\gamma_2} > \max_{0 \neq x \in \mathbb{R}^n} \frac{\langle Dx, x \rangle}{\langle Ax, x \rangle} = \lambda_{\max}(A^{-1}D).$$

This provides a mechanism to determine γ_1 and γ_2 . In applications, $\lambda_{\max}(A^{-1}D)$ can be estimated by using a few steps of the power method. See also section 5 for more on how to select γ_1 and γ_2 .

Using the proposed preconditioner, the system (30), being self-adjoint and positive definite in $\langle \cdot, \cdot \rangle_{\mathcal{H}}$, can be theoretically solved by Krylov's subspace methods in a nonstandard inner product defined by $\mathcal{H}$. In this paper, we solve (30) by the preconditioned CG method in $\langle \cdot, \cdot \rangle_{\mathcal{H}}$, PHCG for short, see section 5.

4. Spectral properties. In this section, spectral properties of the preconditioned matrix $\tilde{\mathcal{A}}$ are studied. Before we start our analysis, we should mention that all eigenvalues of $\tilde{\mathcal{A}}$ are real due to being self-adjoint with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and in what follows, we should show that with selective choices of the parameters γ_1, γ_2 , and α these eigenvalues are positive. Before we state the main theorem, we need the following remark. This remark and its proof can be found in [22].

Remark 4.1. The eigenvalues of

$$B(A + D)^{-1}B^T y = \bar{\lambda} N y$$

are $\{\frac{\sigma_i}{1+\sigma_i}\}_{i=1}^m$, where $\{\sigma_i\}_{i=1}^m$ are the eigenvalues of the matrix $N^{-1}BA^{-1}B^T$.

THEOREM 4.1. *Let T_h be a quasi-uniform mesh and suppose that the positive parameters γ_1, γ_2 , and α in (29) are chosen such that (35) is satisfied with $0 < \gamma_1 < 1$ and $\alpha = \frac{\gamma_1(1-\gamma_1)}{\gamma_2}$. Let $\tilde{\mathcal{A}}$ be the preconditioned matrix in (30). Then $\lambda = \frac{1}{\gamma_1}$ is an eigenvalue of $\tilde{\mathcal{A}}$ with multiplicity n and the remaining m eigenvalues satisfy*

$$(37) \quad \lambda \in \left[\frac{\delta}{1-\gamma_1}, \frac{1}{1-\gamma_1} \right),$$

where $\delta = \frac{c\mu_{\min}\gamma_2}{\gamma_1|K|_{\min} + \gamma_2\mu_{\min}}$ with $\mu_{\min}$ the minimum eigenvalue of the Schur complement $BA^{-1}B^T$, $|K|_{\min}$ the area of the smallest element in T_h , and c a constant independent of h .

Proof. The eigenvalues of $\tilde{\mathcal{A}}$ are the same as those that satisfy the following generalized eigenvalue problem

$$(38) \quad \underbrace{\begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} (\gamma_1 A + \gamma_2 D) & 0 \\ B & -\alpha N \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Since $\mathcal{A}$ is nonsingular, $\lambda \neq 0$. From (38) we get

$$(39) \quad Ax + B^T y = \lambda(\gamma_1 A + \gamma_2 D)x,$$

$$(40) \quad Bx = \lambda Bx - \alpha \lambda N y.$$

It is clear that $x \neq 0$, because if so then by (40) $y = 0$. We claim that 1 cannot be an eigenvalue, because otherwise from (40) $y = 0$ and since $(1 - \gamma_1)A - \gamma_2 D$ is nonsingular, by Remark 3.1, we deduce from (39) that $x = 0$.

Assume that $\lambda = \frac{1}{\gamma_1}$. Then (39) and (40) become

$$(41) \quad B^T y = \frac{\gamma_2}{\gamma_1} D x,$$

$$(42) \quad \left(1 - \frac{1}{\gamma_1}\right) Bx = -\frac{\alpha}{\gamma_1} N y.$$

Substituting for y from (42) into (41) gives

$$\left(\frac{1-\gamma_1}{\alpha}\right) B^T N^{-1} Bx = \frac{\gamma_2}{\gamma_1} Dx.$$

This is true by Lemma 2.1 and because $\alpha = \frac{\gamma_1(1-\gamma_1)}{\gamma_2}$, and since $D \in R^{n \times n}$ is symmetric, $\lambda = \frac{1}{\gamma_1}$ is an eigenvalue of multiplicity n , proving our first assertion.

Now suppose that $\lambda \neq \frac{1}{\gamma_1}$, then we have from (39)

$$B^T y = (\lambda\gamma_1 - 1)Ax + \lambda\gamma_2 Dx = (\gamma_1\lambda - 1)(A + D)x + [(\gamma_2 - \gamma_1)\lambda + 1] Dx;$$

multiplying by $(1 - \lambda)B(A + D)^{-1}$ gives

$$\begin{aligned} (1 - \lambda)B(A + D)^{-1}B^T y &= (1 - \lambda)(\gamma_1\lambda - 1)Bx + (1 - \lambda)[(\gamma_2 - \gamma_1)\lambda + 1] B(A + D)^{-1}Dx, \\ &= (1 - \lambda)(\gamma_1\lambda - 1)Bx + (1 - \lambda)[(\gamma_2 - \gamma_1)\lambda + 1] B(A + D)^{-1}B^T N^{-1}Bx. \end{aligned}$$

But from (40), we have

$$(1 - \lambda)B(A + D)^{-1}B^T y = -\alpha\lambda(\gamma_1\lambda - 1)Ny - \alpha\lambda[(\gamma_2 - \gamma_1)\lambda + 1] B(A + D)^{-1}B^T y.$$

So these eigenvalues satisfy

$$(43) \quad [\alpha(\gamma_2 - \gamma_1)\lambda^2 + (\alpha - 1)\lambda + 1] B(A + D)^{-1}B^T y = -\alpha\lambda(\gamma_1\lambda - 1)Ny.$$

The coefficient of $B(A + D)^{-1}B^T y$ can be factored to give

$$\frac{1}{\gamma_1} (\gamma_1\lambda - 1) [\alpha(\gamma_2 - \gamma_1)\lambda - \gamma_1] B(A + D)^{-1}B^T y = -\alpha(\lambda\gamma_1 - 1)\lambda Ny,$$

and since $\lambda \neq \frac{1}{\gamma_1}$ and $0 < \gamma_1 < 1$, we divide both sides of the last equation by $-\alpha(\lambda\gamma_1 - 1)\lambda$ yielding

$$-\frac{1}{\alpha\gamma_1\lambda} [\alpha(\gamma_2 - \gamma_1)\lambda - \gamma_1] B(A + D)^{-1}B^T y = Ny,$$

which can be written as

$$[B(A + D)^{-1}B^T]^{-1} Ny = \left(\frac{1}{\alpha\lambda} - \frac{\gamma_2}{\gamma_1} + 1\right) y.$$

By Remark 4.1, we have

$$\begin{aligned} \frac{1}{\alpha\lambda_{n+i}} - \frac{\gamma_2}{\gamma_1} + 1 &= \frac{\sigma_i}{1 + \sigma_i}, & i = 1, 2, \dots, m. \\ \lambda_{n+i} &= \frac{\sigma_i\gamma_1}{\alpha(\gamma_1 + \gamma_2\sigma_i)}, & i = 1, 2, \dots, m. \end{aligned}$$

Thus, the remaining m positive eigenvalues lie in the interval

$$\left[\min_i \frac{\sigma_i\gamma_1}{\alpha(\gamma_1 + \gamma_2\sigma_i)}, \max_i \frac{\sigma_i\gamma_1}{\alpha(\gamma_1 + \gamma_2\sigma_i)} \right].$$

Since $\sigma_j > 0$ for all j and the function $y(\sigma) = \frac{\sigma\gamma_1}{\alpha(\gamma_1 + \gamma_2\sigma)}$ is increasing in $[0, +\infty)$ and $y = \frac{1}{1-\gamma_1}$ is a horizontal asymptote, we then have

$$(44) \quad \lambda_{n+i} \in \left[\frac{\sigma_{\min}\gamma_1}{\alpha(\gamma_1 + \gamma_2\sigma_{\min})}, \frac{1}{1-\gamma_1} \right), \quad i = 1, 2, \dots, m,$$

where $\sigma_{\min}$ is the smallest eigenvalue of the matrix $N^{-1}BA^{-1}B^T$. Since the matrix N is diagonal with entries $|K|_1, |K|_2, \dots, |K|_m$, we then have

$$\frac{\mu_{\min}}{|K|_{\max}} \leq \sigma_{\min} \leq \frac{\mu_{\min}}{|K|_{\min}},$$

where $|K|_{\max}$ and $|K|_{\min}$ denote the smallest and largest areas of rectangles in T_h . Now, the left end point in the interval (44) can be bound from below as follows:

$$\begin{aligned} \frac{\sigma_{\min}\gamma_1}{\alpha(\gamma_1 + \gamma_2\sigma_{\min})} &\geq \frac{\frac{\mu_{\min}}{|K|_{\max}}\gamma_1}{\alpha(\gamma_1 + \gamma_2\frac{\mu_{\min}}{|K|_{\min}})} \\ &\geq \frac{|K|_{\min}}{|K|_{\max}} \frac{\mu_{\min}\gamma_1}{\alpha(\gamma_1|K|_{\min} + \gamma_2\mu_{\min})} \\ &\geq \frac{|K|_{\min}}{|K|_{\max}} \frac{\mu_{\min}\gamma_2}{\gamma_1|K|_{\min} + \gamma_2\mu_{\min}} \frac{1}{1-\gamma_1}. \end{aligned}$$

For quasi-uniform T_h , we have

$$\lambda_{n+i} \in \left[\frac{c\mu_{\min}\gamma_2}{\gamma_1|K|_{\min} + \gamma_2\mu_{\min}} \frac{1}{1-\gamma_1}, \frac{1}{1-\gamma_1} \right), \quad i = 1, 2, \dots, m,$$

where the constant c is positive and independent of h . $\square$

Remark 4.2. Theorem 4.1 clearly dictates that α depends on γ_1 and γ_2 . So $P(\gamma_1, \gamma_2, \alpha)$ can be seen as a function of γ_1 and γ_2 . It is therefore convenient to write $P(\gamma_1, \gamma_2)$ instead of $P(\gamma_1, \gamma_2, \alpha)$. We use this notation in what follows.

Remark 4.3. If T_h is a uniform mesh ($c = 1$) and $\gamma = \gamma_2 = \gamma_1$, then $\lambda \in \{\frac{1}{\gamma}\} \cup [\frac{\delta}{1-\gamma}, \frac{1}{1-\gamma})$, where $\delta = \frac{\mu_{\min}}{|K|_{\min} + \mu_{\min}}$.

Remark 4.4. If T_h is a uniform mesh, then for a fixed γ_2 , it is easy to see that δ , in Theorem 4.1, tends to 1 as γ_1 tends to zero. Although this makes the m eigenvalues that belong to the interval $[\frac{\delta}{1-\gamma_1}, \frac{1}{1-\gamma_1})$ tightly clustered around 1, the condition number, on the other hand, becomes extremely large due to the fact that the other n eigenvalues are $\frac{1}{\gamma_1}$. A better alternative is clearly to push δ towards 1 by making γ_2 large enough. In fact, we can select γ_2 in a precise way to make δ as close as we wish to 1. To see this, assume that we want $\delta = 1 - \varsigma$, where $\varsigma \in (0, 1)$. Then solving for γ_2 gives

$$(45) \quad \gamma_2 = \left(\frac{1}{\varsigma} - 1 \right) \frac{|K|_{\min}}{\mu_{\min}} \gamma_1.$$

Now one can set $\gamma_1 = \frac{1}{2}$ to have $2 = \frac{1}{\gamma_1} = \frac{1}{1-\gamma_1}$ and take γ_2 large enough to drive the left bound $\frac{\delta}{1-\gamma_1}$ to 2. Such a pair of γ_1 and γ_2 is optimal in that it results in a very tight clustering of the eigenvalues of the preconditioned system.

Remark 4.5. We have seen in Remark 3.2 that in order for $\mathcal{H}$ to be positive definite, γ_1 and γ_2 must satisfy the inequality (36). Obviously an optimal choice for good clustering, i.e., $\gamma_1 = 1/2$ and γ_2 large enough, will violate this last inequality. To strike a balance between having a perfect clustering and a positive definite $\mathcal{H}$, we start by selecting γ_1 very small, which makes δ tend to 1 and therefore we have a tight interval $[\frac{\delta}{1-\gamma_1}, \frac{1}{1-\gamma_1})$, and choosing γ_2 such that

$$\gamma_2 < \frac{1 - \gamma_1}{\lambda_{\max}(A^{-1}D)}$$

(see the discussion at the end of section 5 on practical ways of choosing γ_2).

5. PHCG METHOD. We start by giving Algorithm 1 for the Hestenes and Stiefel [10] CG method in a nonstandard inner product $\langle \cdot, \cdot \rangle_H$. This algorithm computes iterates such that $\|x - x_k\|_H$ is minimized. Since $HP^{-1}\mathcal{A}$ is symmetric and positive definite, then $\|\cdot\|_{HP^{-1}\mathcal{A}}$ defines a norm.

Algorithm 1: PHCG.

Input: $\mathcal{A} \in \mathbb{R}^{(n+m) \times (m+n)}$, $b \in \mathbb{R}^{n+m}$.
Output: $\min \|x - x_k\|_H$.
Let: $k = 0$
Choose: x_k, tol
Compute: $r_k = P^{-1}(b - \mathcal{A}x_k)$
Set: $p_k = r_k$
while $\|r_k\|_H > \text{tol}$ **do**
 $k = k + 1$
 if $k = 1$ **then**
 $p_1 = r_0$
 else
 $\beta_k = \langle r_{k-1}, r_{k-1} \rangle_H / \langle r_{k-2}, r_{k-2} \rangle_H$
 $p_k = r_{k-1} + \beta_k p_{k-1}$
 end
 $\alpha_k = \langle r_{k-1}, r_{k-1} \rangle_H / \langle P^{-1}\mathcal{A}p_k, p_k \rangle_H$
 $x_k = x_{k-1} + \alpha_k p_k$
 $r_k = r_{k-1} - \alpha_k P^{-1}\mathcal{A}p_k$
end
 $x = x_k$

The attractive property of the Hestenes and Stiefel CG method is that the method minimizes $\| \text{error} \|_{\mathcal{A}}$ at each step, where $\mathcal{A}$ is the coefficient matrix of the linear system. The PHCG method in Algorithm 1 requires that the preconditioned matrix be self-adjoint and positive definite with respect to the inner product $\langle \cdot, \cdot \rangle_H$. The PHCG method minimizes the error at each step but in the $\|\cdot\|_{HP^{-1}\mathcal{A}}$ -norm. This property is summarized in the following lemma [21].

LEMMA 5.1 (see [21]). *The error, e_k , of the k th PHCG iterate is given by*

$$\frac{\|e_k\|_{HP^{-1}\mathcal{A}}}{\|e_0\|_{HP^{-1}\mathcal{A}}} \leq \min_{p \in \Pi_{k,p(0)=1}} \max_{\lambda \in \sigma(P^{-1}\mathcal{A})} |p(\lambda)|,$$

where $\Pi_{k,p(0)=1}$ is the set of real polynomials of degree k with the constant term equal to 1 and $\sigma(P^{-1}\mathcal{A})$ is the spectrum of the matrix $P^{-1}\mathcal{A}$.

The convergence of the PHCG is guaranteed whenever $\mathcal{H}$ is positive definite. Remark 3.2 gives a lower bound that helps find the parameters γ_1 and γ_2 that ensure the positive definiteness of $\mathcal{H}$. However, it may not be very practical in the sense that one needs to find the maximum eigenvalue of the matrix $A^{-1}D$. So we describe in what follows a more practical technique. We approximate $\lambda_{\max}(A^{-1}D)$ in inequality (36) by using some bounds on $\lambda_{\max}(A^{-1}D)$ in terms of the extremal eigenvalues of the element matrices that are cheaper to calculate. In this paper we are considering Raviart–Thomas finite element discretization; see subsection 2.1. So let A^e and D^e be the element mass and divergence matrices, respectively. We now proceed to state the following lemma.

LEMMA 5.2. *Let $\lambda_{\min}(A)$ and $\lambda_{\min}(D)$ be the minimal eigenvalues of A and D , respectively, and let $\lambda_{\max}(A)$ and $\lambda_{\max}(D)$ be the maximal eigenvalues of A and D , respectively. Denote the minimal and maximal eigenvalues of the element matrices, A^e and D^e , by $\lambda_{\min}(A^e)$, $\lambda_{\min}(D^e)$, $\lambda_{\max}(A^e)$, and $\lambda_{\max}(D^e)$. Then*

$$(46) \quad \lambda_{\max}(A^e) \leq \lambda_{\max}(A) \leq 2\lambda_{\max}(A^e),$$

$$(47) \quad \lambda_{\max}(D^e) \leq \lambda_{\max}(D) \leq 2\lambda_{\max}(D^e),$$

and

$$(48) \quad \lambda_{\min}(A^e) \leq \lambda_{\min}(A),$$

$$(49) \quad \lambda_{\min}(D^e) \leq \lambda_{\min}(D).$$

Proof. The proof utilizes element-by-element analysis and is similar to that of Theorem 1 in [9], so we omit it. $\square$

Remark 5.1. The extremal eigenvalues of the element matrices are easy to calculate. In fact, we have

$$\lambda_{\min}(A^e) = \frac{h^2}{6}, \quad \lambda_{\min}(D^e) = 0, \quad \lambda_{\max}(A^e) = \frac{h^2}{4}, \quad \lambda_{\max}(D^e) = 4,$$

where h is the mesh size. So by Lemma 5.2, we have

$$0 = \frac{\lambda_{\min}(D^e)}{2\lambda_{\max}(A)^e} \leq \frac{\lambda_{\min}(D)}{\lambda_{\max}(A)} \leq \lambda_{\max}(A^{-1}D) \leq \frac{\lambda_{\max}(D)}{\lambda_{\min}(A)} \leq \frac{2\lambda_{\max}(D^e)}{\lambda_{\min}(A)^e} = \frac{48}{h^2}.$$

So one may approximate $\lambda_{\max}(A^{-1}D)$ by $\frac{48}{h^2}$.

We shall describe an approach to obtain the parameters γ_1 and γ_2 which give a positive definite matrix $\mathcal{H}$. In Remark 3.2, γ_1 and γ_2 are required to satisfy (36). We propose two techniques to handle this inequality. The first technique is to approximate $\lambda_{\max}(A^{-1}D)$ by applying a few iterations of the power method. When using the power method the iteration equation $w_{r+1} = A^{-1}Dw_r$ is replaced by solving the linear system

$$(50) \quad Aw_{r+1} = Dw_r.$$

In the case $K^{-1} = I$, the matrix A is the mass matrix. Several efficient linear solvers are available for (50). The second technique is to derive an upper bound for $\lambda_{\max}(A^{-1}D)$. In the following, we will prove that

$$\lambda_{\max}(A^{-1}D) \leq \frac{48}{\hat{\gamma}h^2},$$

where $\hat{\gamma}$ is given in (8). We begin by stating the following lemma which gives bounds on the eigenvalues of the matrices A and D . This lemma can be proved by applying a simple element-by-element analysis [9, 11, 30].

LEMMA 5.3. *Let the smallest eigenvalue of A and largest eigenvalue of D be denoted by $\lambda_{\min}(A)$ and $\lambda_{\max}(D)$, respectively. Then*

$$(51) \quad \max_e \{\lambda_{\max}(D^e)\} \leq \lambda_{\max}(D) \leq 2 \max_e \{\lambda_{\max}(D^e)\}$$

and

$$(52) \quad \min \{\lambda_{\min}(A^e)\} \leq \lambda_{\min}(A),$$

where $\lambda_{\max}(D^e)$ represents the largest eigenvalue of the corresponding element matrix D^e and $\lambda_{\min}(A^e)$ represents the smallest eigenvalue of the corresponding element matrix A^e .

Proof. For a uniform mesh the element divergence matrix D^e and the element matrix A^e of the Raviart–Thomas element with lowest degree are

$$D^e = \begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix} \quad \text{and} \quad A^e = \frac{h^2}{4} \begin{bmatrix} a & 0 & -b & 0 \\ 0 & a & 0 & -b \\ -b & 0 & a & 0 \\ 0 & -b & 0 & a \end{bmatrix},$$

where

$$a = \left(\frac{1}{3} + \frac{1}{2\sqrt{3}} \right) (K^{-1}(p_1) + K^{-1}(p_2)) + \left(\frac{1}{3} - \frac{1}{2\sqrt{3}} \right) (K^{-1}(p_3) + K^{-1}(p_4)),$$

$$b = -\frac{1}{6} (K^{-1}(p_1) + K^{-1}(p_2) + K^{-1}(p_3) + K^{-1}(p_4)).$$

The matrix A^e is constructed by applying four-point quadrature over a square domain in two dimensions. The quadrature points are p_1, p_2, p_3 , and p_4 . In the case $K^{-1} = I$ the element matrix A^e becomes

$$A^e = h^2 \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{6} & 0 \\ 0 & \frac{1}{3} & 0 & -\frac{1}{6} \\ -\frac{1}{6} & 0 & \frac{1}{3} & 0 \\ 0 & -\frac{1}{6} & 0 & \frac{1}{3} \end{bmatrix}.$$

The extremal eigenvalues of the element matrices D^e and A^e are

$$(53) \quad \lambda_{\max}(D^e) = 4,$$

$$(54) \quad \lambda_{\min}(A^e) = \frac{h^2}{4} \left[\left(\frac{1}{6} + \frac{1}{2\sqrt{3}} \right) (K^{-1}(p_1) + K^{-1}(p_2)) + \left(\frac{1}{6} - \frac{1}{2\sqrt{3}} \right) (K^{-1}(p_3) + K^{-1}(p_4)) \right].$$

Equations (51) and (52) yield

$$(55) \quad \lambda_{\max}(D) \leq b,$$

$$(56) \quad \lambda_{\min}(A) \geq \frac{\hat{\gamma}h^2}{6}.$$

Equations (55) and (56) give

$$\lambda_{\max}(A^{-1}D) \leq \frac{48}{\hat{\gamma}h^2}.$$

□

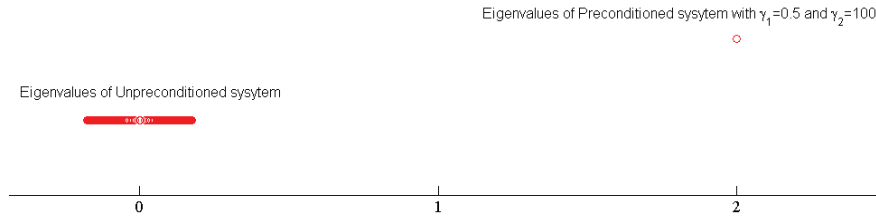


FIG. 1. Eigenvalues of the system without preconditioning (left). Eigenvalues of the system with preconditioning $\gamma_1 = 0.5$ and $\gamma_2 = 100$ (right).

TABLE 1
Eigenvalues of preconditioned matrix with uniform mesh.

h	$\mu_{\min}$	$\lambda_1 = 1/\gamma_1$ (frequency)	Remaining m eigenvalues		Lower bound $\frac{\delta}{1-\gamma_1}$	Upper bound $\frac{1}{1-\gamma_1}$
			min	max		
$\frac{1}{8}$	0.00060	100 (144)	1.009540	1.010094	1.009540	1.010101
$\frac{1}{16}$	0.00014	100 (544)	1.009535	1.010099	1.009535	1.010101
$\frac{1}{32}$	0.00003	100 (2112)	1.009533	1.0101006	1.009533	1.0101010

6. Numerical examples. In this section, we study (1) on $\Omega = [0, 1] \times [0, 1]$ using rectangular elements. In subsection 6.1, we investigate the eigenvalues of the preconditioned system. A comparison study between P_{div} and $P(\gamma_1, \gamma_2)$ is presented in subsection 6.2. We solve the preconditioned system by GMRES using three different preconditioners in subsection 6.3.

In all of our numerical experiments, the action of P^{-1} on the upper left block matrix is approximated by the aggregation-based algebraic Multigrid method, AGMG, proposed by Notay [17], [18], and Napov and Notay [16]. AGMG uses a Krylov subspace method as smoother at every level. All default parameters of AGMG were used. All experiments were carried out using a PC, Intel Core i5-2500 CPU @3.30 Ghz, 4 GB memory (RAM), Windows 10 Professional (64 bit) operating system, and MATLAB R2015b.

6.1. Eigenvalues clustering. We first consider (1) on $\Omega = [0, 1] \times [0, 1]$, $K = I$, and $\partial\Omega = \partial\Omega_D$. With $h = 1/16$, we start by comparing the clustering of the eigenvalues of the system (2) without preconditioning against the eigenvalues of the same system after preconditioning using an optimal pair of parameters, as suggested by Remark 4.4, $\gamma_1 = 0.5$ and $\gamma_2 = 100$. This comparison is shown in Figure 1.

The eigenvalues of the preconditioned system $\lambda_1, \dots, \lambda_{n+m}$ are reported in Table 1 for different uniform mesh sizes. We choose the parameters $(\gamma_1, \gamma_2, \alpha) = (0.01, 0.9, 0.011)$ to show, in the last four columns, that the bounds in Theorem 4.1 are tight.

In Figure 2, we illustrate how the parameters γ_1 and γ_2 control the clustering behavior of the eigenvalues of the preconditioned matrix. With $h = 1/32$, six different values of γ_1 and γ_2 are selected. The first five values are random, with γ_1 restricted to be less than or equal to $1/2$ so that the upper limit of the eigenvalues is less than or equal to 2 and for the last one we choose an optimal pair of $\gamma_1 = 0.5$ and large enough γ_2 (see Remark 4.4). The n eigenvalues ($1/\gamma_1$) are not plotted. We only plot the remaining m eigenvalues in different levels bottom to top; each level corresponds to a different pair of γ_1 and γ_2 . The optimal pair is the one on the top level where all eigenvalues are clustered around 2.

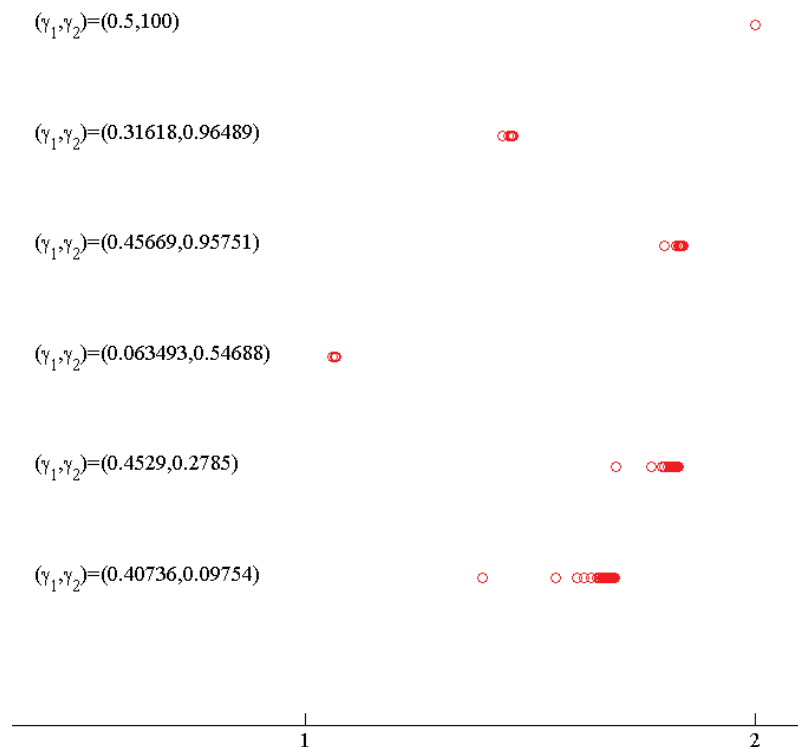


FIG. 2. Eigenvalues clustering with different values of γ_1 and γ_2 and $h = 1/32$. The first five bottom to top are for random γ_1 and γ_2 . The top is for an optimal pair of $(\gamma_1, \gamma_2) = (0.5, 4000)$.

For the calculations above, uniform meshes were used with $K(x) = I$. However, a permeability of small values on the diagonal is more practical. So we consider

$$(57) \quad K(x) = \begin{cases} \epsilon I & \text{for all } x \in \Omega_0, \\ I & \text{for all } x \in \Omega \setminus \Omega_0, \end{cases}$$

where I is the identity matrix, $\Omega_0 = [0.25, 0.5] \times [0.25, 0.75] \subset \Omega$, and $\epsilon \in [10^{-6}, 10^6]$. The corresponding values of the minimum eigenvalue, $\mu_{\min}$, of the Schur complement $BA^{-1}B^T$ are reported in [22, Table 2.3]. For very small values of $\epsilon \ll 1$, $\mu_{\min}$ tends to zero. This would force the left-hand bound of the m eigenvalues in Theorem 4.1 to converge to 0. In [22, Figure 2.2 (left)] it was illustrated that a subset of the eigenvalues of the preconditioned system associated with (57) decay. The presence of γ_2 in our bound can help overcome this undesirable behavior by taking γ_2 large enough. In fact we select γ_2 in an adaptive manner with $\gamma_2 = 100/\epsilon$ for $\epsilon \in [10^{-6}, 10^{-1}]$. Using this approach all eigenvalues of the preconditioned system associated with (57) are tightly clustered around 2 (see Figure 3 (left)). If $\epsilon \in [1, 10^6]$, we select $\gamma_1 = 0.5$ and a fixed $\gamma_2 = 100$ (see Figure 3 (right)).

6.2. Comparisons with P_{div} . In this subsection we consider an example to compare the proposed preconditioner $P(\gamma_1, \gamma_2)$ with the block-diagonal preconditioner

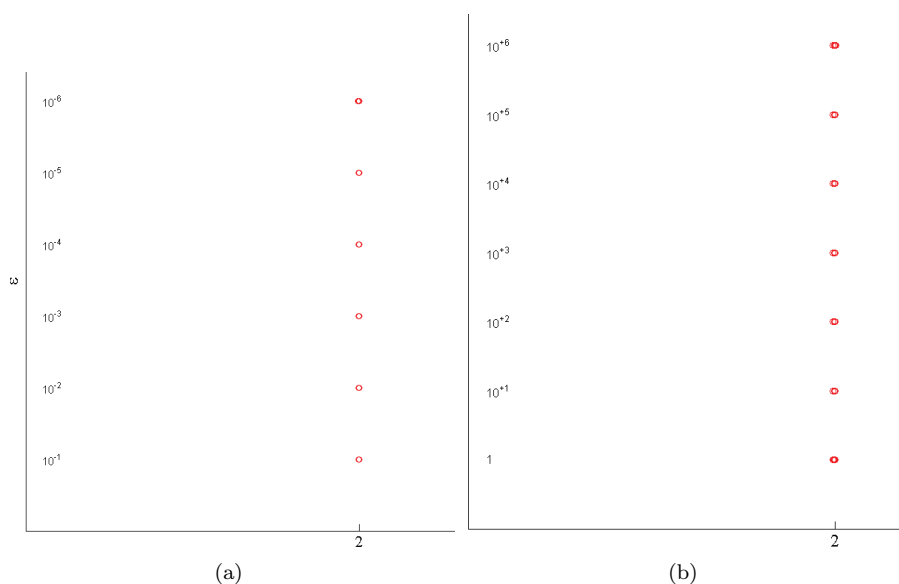


FIG. 3. Eigenvalues of preconditioned system with $h = 1/32$ and discontinuous permeability $\epsilon \in [10^{-1}, 10^{-6}]$, $\gamma_1 = 0.5$, and an adaptive γ_2 (left) $\epsilon \in [1, 10^6]$, $\gamma_1 = 0.5$, and $\gamma_2 = 100$ (right).

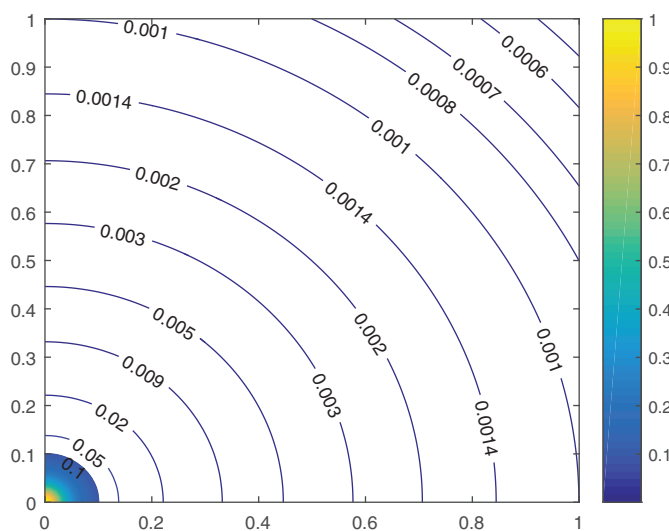


FIG. 4. Contour of the function $\frac{1}{1+c(x^2+y^2)}$ with $c = 1000$.

P_{div} . We take K to be

$$K(x, y) = \begin{bmatrix} \frac{1}{1+c(x^2+y^2)} & 0 \\ 0 & \frac{1}{1+c(x^2+y^2)} \end{bmatrix}.$$

For large values of c this is a challenging problem [22]. The function $K(x, y)$ is shown in Figure 4. We apply PHCG with several set of parameters. For the P_{div} preconditioner

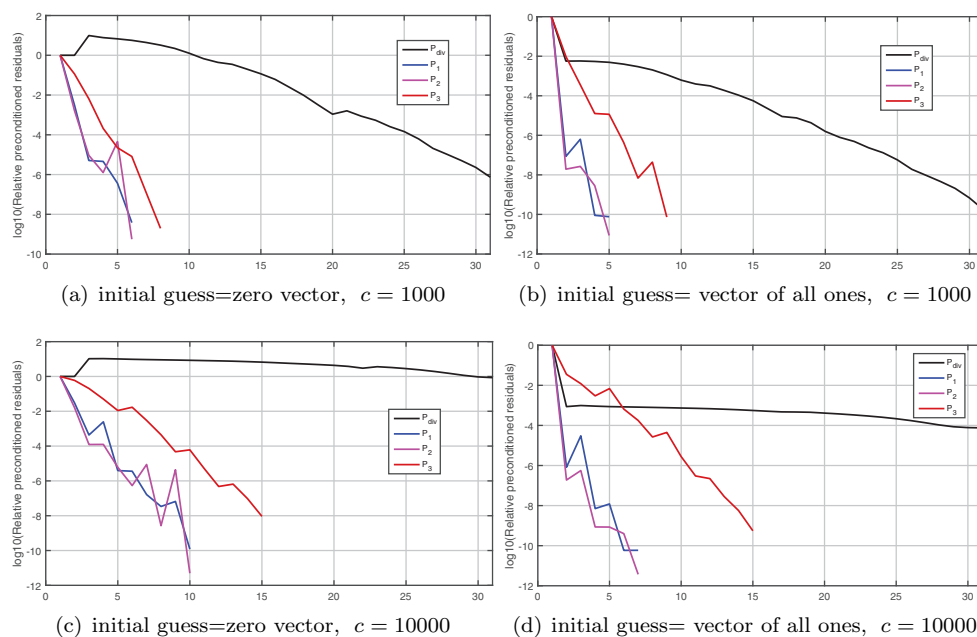


FIG. 5. The left-hand plots show the preconditioned relative residual plots for P_1 (blue line), P_2 (magenta line), P_3 (red line), and PMINRES for P_{div} (black line) with zero vector as an initial guess. The right-hand plots show the preconditioned relative residual plots for P_1 (blue line), P_2 (magenta line), P_3 (red line), and PMINRES for P_{div} (black line) with the vector of all ones as an initial guess.

we use the built-in MATLAB minres function, which is based on the MINRES method of Paige and Saunders [19]. For PHCG, the stopping criterion is a reduction in the norm of the relative residuals to 10^{-10} , measured in the $\|\cdot\|_{P^{-1}}$ norm with $P = P_{(\gamma_1, \gamma_2)}$. For MINRES, the stopping criterion is a reduction in the norm of the relative residuals to 10^{-10} , measured in the $\|\cdot\|_{P^{-1}}$ norm with $P = P_{div}$. In this example, we have used two initial guesses: the zero vector and the vector of all ones with two different values for c . The cases are: $P_1 = P(6.7 \times 10^{-9}, 2.8 \times 10^{-5})$ and $P_2 = P(1.5 \times 10^{-9}, 1.19 \times 10^{-5})$. Figures 5(a)–5(d) show the plot of the log10 of the relative preconditioned residual versus iteration number. We note that P_1 and P_2 converge faster than P_{div} . We should note that we solved the system using the MATLAB backslash operator. Then we computed the infinity norm of the difference between the MATLAB solution and the solutions using the preconditioners P_{div} , P_1 , and P_2 . The difference was less than 10^{-7} in all cases. However, the number of iterations in the P_{div} case was at least three times that in the P_1 and P_2 cases. Note that the selection of γ_1 and γ_2 in P_1 and P_2 satisfy the positive definiteness condition (36). We observed that P_{div} is sensitive to the selection of initial guess vector (see Figures 5(a) and 5(c)). This was further checked and confirmed by using several random initial guess vectors generated by the built-in MATLAB command rand. When we run the code with selections of γ_1 and γ_2 that do not satisfy (36), some of them fail to converge and some converge such as $P_3 = P(0.0009, 0.09)$ which is shown in the figures.

In Figures 5(a) and 5(c), we can notice that although there is an overall reduction in the residual, there are cases where a sudden increase occurs followed by a reduction; for example, P_{div} starts with an increase followed by a decrease, whereas P_2 has few increases along the curve followed by decreases. The behavior of these curves is caused by the difficulty of this example. The permeability has a high variation

TABLE 2
Zero vector and $c = 1000$.

	P_1	P_2	P_3
$\mu_{\min}$	8.2489616×10^{-7}	8.2489616×10^{-7}	8.2489616×10^{-7}
δ	0.93386354	0.96403518	0.25254739
Lower bound	0.93386354	0.96403518	0.25277489
Upper bound	1.000000007	1.000000002	1.000900811
Interval length	0.066136463	0.035964825	0.74812592
Isolated λ	1.4925373×10^8	6.6666667×10^8	1111.1111
$\frac{1-\gamma_1}{\gamma_2}$	35714.285	84033.613	11.101111
$\lambda_{\max}$ in (36)	62650.901	62650.901	62650.901

TABLE 3
Vector of all ones and $c = 1000$.

	P_1	P_2	P_3
$\mu_{\min}$	8.2340362×10^{-6}	8.2340362×10^{-6}	8.2340362×10^{-6}
δ	0.99295512	0.9962765	0.77130632
Lower bound	0.99295513	0.9962765	0.77200112
Upper bound	1.000000007	1.000000002	1.000900811
Interval length	0.0070448812	0.0037235009	0.22889969
Isolated λ	1.4925373×10^8	6.6666667×10^8	1111.1111
$\frac{1-\gamma_1}{\gamma_2}$	35714.285	84033.613	11.101111
$\lambda_{\max}$ in (36)	62649.615	62649.615	62649.615

on some regions of the domain which is the reason behind the difficulty. Also, the permeability is very low in a wide range of the domain as shown in Figure 4. Hence γ_1 and γ_2 were chosen to be small values. We expect better results if the values of γ_1 and γ_2 vary and depend on the varying permeability. The values of all the parameters appearing in Theorem 4.1 and inequality (36) are recorded in Tables 2–5.

6.3. GMRES. The preconditioned system we are considering is not symmetric which makes GMRES a method of choice to solve it. In this subsection, we solve the preconditioned system using GMRES and report on its convergence in terms of the number of iterations. The stopping tolerance is 10^{-9} .

In this example, we let $K = I$ and $f = 1$. The number of iterations are shown in Table 6 for solving the unpreconditioned system ($P = I$), and preconditioned with $P = P(0.9, 0.001)$ and $P = P(0.5, 900)$ which is an optimal pair. Table 6 shows that the optimal pair gives the least number of iterations.

TABLE 4
Zero vector and $c = 10000$.

	P_1	P_2	P_3
$\mu_{\min}$	8.2489616×10^{-7}	8.2489616×10^{-7}	8.2489616×10^{-7}
δ	0.93386354	0.96403518	0.25254739
Lower bound	0.93386354	0.96403518	0.25277489
Upper bound	1.000000007	1.000000002	1.000900811
Interval length	0.066136463	0.035964825	0.74812592
Isolated λ	1.4925373×10^8	6.6666667×10^8	1111.1111
$\frac{1-\gamma_1}{\gamma_2}$	35714.285	84033.613	11.101111
$\lambda_{\max}$ in (36)	32102.205	32102.205	32102.205

TABLE 5
Vector of all ones and $c = 10000$.

	P_1	P_2	P_3
$\mu_{\min}$	8.2489616×10^{-7}	8.2489616×10^{-7}	8.2489616×10^{-7}
δ	0.93386354	0.96403518	0.25254739
Lower bound	0.93386354	0.96403518	0.25277489
Upper bound	1.000000007	1.000000002	1.000900811
Interval length	0.066136463	0.035964825	0.74812592
Isolated λ	1.4925373×10^8	6.6666667×10^8	1111.1111
$\frac{1-\gamma_1}{\gamma_2}$	35714.285	84033.613	11.101111
$\lambda_{\max}$ in (36)	32102.204	32102.204	32102.204

TABLE 6
GMRES iterations.

h	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$
$P = I$	101	243	497	983
$P = P(0.9, 0.001)$	22	30	32	33
$P = P(0.5, 900)$	3	3	3	4

7. Summary. Block-triangular preconditioners with two parameters for the linear system obtained from the lowest order Raviart–Thomas mixed finite element discretization of Darcy’s equations are presented. The preconditioned system is nonsym-

metric but it is self-adjoint in a nonstandard inner product. Eigenvalues analysis and eigenvalues bounds are obtained and verified with several numerical computations. A small number of iterations are observed when the preconditioned CG method is used.

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